CS174 Spr 99 Lecture 21 Summary John Canny

Number Theory

Divisibility: We use the notation

$$a|b$$

to mean "a divides b" exactly. So $3|6$, $15|45$ etc. while the symbol $\not|$ means "does not divide evenly", so $5 \not| 8$, and $12 \not| 16$ etc. Note that $a|0$ holds for all a, i.e. zero is divisible by everything.

Greatest Common Divisor The greatest common divisor (GCD) of a and b is defined as:

$$gcd(a, b) = \max\{g :\ g|a \text{ and } g|b\}$$

So for example, gcd(20,65) = 5, gcd(19,38) = 19, gcd(5,12) = 1 etc.

Lowest Common Multiple The lowest common multiple (LCM) of a and b is defined as:

$$lcm(a, b) = \min\{l :\ a|l \text{ and } b|l\}$$

So for example, lcm(20,30) = 60, lcm(19,38) = 38, gcd(5,12) = 60 etc.

Lemma The GCD $g(a, b)$ and LCM $l(a, b)$ satisfy:

$$lcm(a, b) = \frac{ab}{gcd(a,b)}$$

Proof If g is the GCD of a and b then they can both be written:

$$a = ga'$$

$$b = gb'$$

and substituting into $l = ab/g$ gives $l = a'gb'$. That form of l is clearly divisible by both a and b and so it is a common multiple.

Then either $l = ab/g$ is the LCM or there is another common multiple l_0 which is smaller. Suppose a smaller LCM l_0 exists. Now the LCM divides all common multiples of a and b, and in particular, it divides ab. Define

$$g_0 = ab/l_0$$

The ratio l_0/b is an integer (l_0 is a multiple of b), and rearranging the last equation shows that g_0 is a factor of a:

$$g_0(l_0/b) = a$$

Similarly the ratio l_0/a is an integer, and rearranging shows that g_0 is also a factor of b.

Thus g_0 is a common multiple of a and b. Now since $l = ab/g$ and $l_0 = ab/g_0$ and $l > l_0$, we must have $g_0 > g$. But that is a contradiction (g is the greatest common divisor of a and b).

Factorization

A *prime* $p > 1$ is a number with no divisors except for p and 1.

Other numbers are called *composite*. Composite numbers have unique factorizations as powers of primes. That is, every number (primes too) n can be uniquely expressed as a product:

$$n = p_1^{e_1} \cdots p_k^{e_k}$$

For example $84 = 2^2 \times 3 \times 7$.

Division Theorem

Given a dividend a and a divisor b, there are unique integers q and $r \in [0, ..., b-1]$ such that:

$$a \;=\; qb + r$$

and we write $q = a \operatorname{div} \; b$ for the quotient and $r = a \bmod b$ for the remainder.

Note that $b|a$ is equivalent to $a \bmod b = 0$

With the results we have so far, we can prove an interesting theorem. You already know this theorem is true, but nevertheless its useful to know one reason why.

Theorem: There are infinitely many primes.

Proof: Suppose not. Then there are finitely many primes, $p_1, \ldots, p_N$. Lets define a new number Q as

$$Q \;=\; p_1 \cdots p_N + 1$$

By the unique factorization theorem, this number must be a product of powers of primes. But notice that:

$$\begin{array}{rclcc} Q \bmod p_1 & = & 1 & \text{so} & p_1 \not| Q \\ Q \bmod p_1 & = & 1 & \text{so} & p_1 \not| Q \\ \vdots & & & \vdots & \vdots \\ Q \bmod p_N & = & 1 & \text{so} & p_N \not| Q \end{array}$$

In other words, Q is not divisible by any of the p_is, which means it has no prime factors at all. That is, it must be equal to 1. But that is a contradiction because all the p_i's are > 1 and so is their product. QED

Euclid's Algorithm

Euclid's algorithm is a method for efficiently computing GCDs. Its based on the observation that if

$$g \;=\; \gcd(a, b)$$

then

$$r \;=\; a \bmod b \;=\; a - qb$$

is also divisible by g because both a and b are. By repeatedly taking remainders, we can reduce the size of the numbers whose gcd we are computing, until eventually we get the gcd itself. Let

$$\begin{aligned}
r_1 &= a \\
r_2 &= b \\
r_3 &= r_1 \bmod r_2 \\
r_4 &= r_2 \bmod r_3 \\
\vdots \;&\vdots\; \vdots \\
r_k &= r_{k-2} \bmod r_{k-1} \\
0 &= r_{k-1} \bmod r_k
\end{aligned}$$

We will say more in a moment about why this must terminate with a zero. To see that this computes the GCD, we notice that common divisors of any pair of consecutive r_i's are the same. That is

Lemma
The number g is a common divisor of r_j and r_{j-1} if and only if it is a common divisor of r_j and r_{j+1}.

Proof
First suppose g is a common divisor of r_j and r_{j-1}. Then r_{j+1} can be written

$$r_{j+1} \;=\; r_{j-1} (\bmod\ r_j) \;=\; r_{j-1} - q_{j+1} r_j$$

where q_j is the j^{th} quotient. Clearly, any common divisor of r_{j-1} and r_j will be a divisor of r_{j+1}. If we simply rearrange this expression:

$$r_{j+1} + q_{j+1} r_j \;=\; r_{j-1}$$

then we can see that any divisor of both r_{j+1} and r_j is also a divisor of r_{j-1}. QED

It follows immediately that very last pair in the sequence is $r_k, 0$, so $\gcd(r_{k-1}, r_k) = \gcd(r_k, 0) = r_k$. We can apply the equality $\gcd(r_{j-1}, r_j) \;=\; \gcd(r_j, r_{j+1})$ all the way back to the begining to conclude that:

$$\gcd(a, b) \;=\; \gcd(r_1, r_2) \;=\; \gcd(r_2, r_3) \;= \cdots = \gcd(r_k, 0) \;=\; r_k$$

That is, $gcd(a, b) \;=\; r_k$, the last non-zero remainder in the sequence.

Next we consider the speed of convergence. First of all, notice that if $a > 0$ and $b > 0$, then the sequence is strictly decreasing. So it must eventually terminate with a zero. How fast does it decrease?

We show that if the algorithm runs for k steps (k defined as above), then

Lemma
If Euclid's algorithm runs for k steps, then

$$\begin{aligned} a &\geq F_{k+1} \\ b &\geq F_k \end{aligned}$$

where F_k is the k^{th} Fibonacci number.

Proof
Assume wlog that $a > b > 0$ (else swap a, b). Then at every step there is a positive quotient $q_j > 0$ such that

$$r_j = r_{j-2} - q_j r_{j-1}$$

where $q_j > 0$ because $q_j = r_{j-2} \mathrm{div}\ \ r_{j-1}$ and $r_{j-2} > r_{j-1}$.

The proof is by induction. Assume: $r_{k-j+2} \geq F_j$

The base cases are $j = 2, 3$, where $F_2 = 1$, $F_3 = 2$. The corresponding remainders are r_k and r_{k-1}. Now $r_k = \gcd(a, b) \geq 1$, and $r_{k-1} > r_k \geq 2$ since the sequence is decreasing. This completes the case cases.

For the inductive step, we use the identity:

$$r_{i-2} = r_i + q_i r_{i-1}$$

Making the substitution $i = k - j + 2$ gives:

$$r_{k-j} = r_{k-j+2} + q_i r_{k-j+1}$$

And we substitute using $r_{k-j+2} \geq F_j$, giving:

$$r_{k-j} \geq F_j + q_i F_{j+1} \geq F_{j+2} \qquad \text{remember } \ q_i > 0$$

So we have moved the Fibonacci inequality up one level in the sequence. We get the results we are looking for by setting $j = k - 1$ for a and $j = k - 2$ for b:

$$\begin{aligned} a &= r_1 \geq F_{k+1} \\ b &= r_2 \geq F_k \end{aligned}$$

Which completes the proof. QED

From this result it follows that if $a < F_{k+1}$, the algorihm must run for less than k step. Now the Fibonacci numbers satisfy $F_k \approx \gamma^k$ where γ is the golden ratio $(1 + \sqrt{5})/2$. Thus k is at most about $\log_\gamma(a)$. So we can say that:

Corollary Euclid's algorithm takes $O(\log a)$ steps to compute $\gcd(a, b)$.

Extended Euclid

We can get more information from Euclid's algorithm by doing some book-keeping. In particular, if $g = \gcd(a, b)$, we will compute x, y such that

$$g \;=\; ax + by$$

This follows easily by induction. Suppose that we can express

$$r_i = x_i a + y_i b$$

Clearly this is true for $i = 1, 2$ with the (x_i, y_i) pairs $(1, 0)$ and $(0, 1)$ respectively. Suppose its true for i and $i + 1$. We prove it holds for $i + 2$. Now

$$r_{i+2} \;=\; r_i - q_{i+2} r_{i+1} \;=\; (ax_i + by_i) + q_{i+2}(ax_{i+1} + by_{i+1}) \;=\; a(x_i + q_{i+2}x_{i+1}) + b(y_i + q_{i+2}y_{i+1})$$

Which proves the identity we were looking for and establishes the inductive formulae:

$$\begin{aligned} x_{i+2} &= (x_i + q_{i+2}x_{i+1}) \\ y_{i+2} &= (y_i + q_{i+2}y_{i+1}) \end{aligned}$$

And if r_k is the last remainder in the sequence, we see that

$$g \;=\; gcd(a, b) \;=\; r_k \;=\; x_k a + y_k b$$

To implement extended Euclid, simply initialize (x_1, y_1) and (x_2, y_2) to $(1, 0)$ and $(0, 1)$ respecively, and use the above inductive formula as the remainders are computed.